

# Statistics — Level 1

## FAANG / MAANG Data Science & AI/ML Interview Guide

Intuition first → reasoning → formula → calculation → interview application.

### 1. Probability Foundations

Probability measures how likely an event is: 0 to 1 (0% to 100%).

Sample space = all possible outcomes. An event = the outcome(s) we care about.

Complement:  $P(\text{not } A) = 1 - P(A)$ . For “at least one,” use  $P(\text{at least one}) = 1 - P(\text{none})$ .

Independence means one event does not change another event’s probability. For independent events,  $P(A \text{ and } B) = P(A)P(B)$ .

Conditional probability  $P(A|B)$  means probability of A after restricting attention to cases where B is true:  $P(A|B) = P(A \text{ and } B) / P(B)$ .

**Memory hook:** Complement = opposite. Conditional = shrink the world to the “given” group.

### 2. Bayes & Classifier Thinking

Bayes:  $P(A|B) = P(B|A)P(A) / P(B)$ . It updates belief after seeing evidence.

Spam example: 10,000 emails contain 100 spam and 9,900 legitimate. If recall is 90%, 90 spam emails are flagged. If false-positive rate is 5%, 495 legitimate emails are flagged.

Total flagged = 585. Precision =  $90 / 585 \approx 15.38\%$ . Recall =  $90 / 100 = 90\%$ . False-positive rate =  $495 / 9,900 = 5\%$ .

Key lesson: when positives are rare, even a small false-positive rate can create many false alarms.

**Memory hook:** “Given flagged” means denominator = all flagged. “Given actually spam” means denominator = all spam.

### 3. Random Variables

A random variable assigns a number to a random outcome. Example:  $X$  = number of purchases by a user.

Discrete variables are countable: clicks, purchases, defects, heads. Continuous variables are measured on a continuum: height, latency, temperature, time.

**Memory hook:** Discrete = count it. Continuous = measure it.

### 4. Bernoulli

A Bernoulli trial has exactly two outcomes: success/failure or 1/0.

Example: click = 1 and no click = 0. If success probability is  $p$ , failure probability is  $1 - p$ .

For Bernoulli  $X$ :  $E[X] = p$  and  $\text{Var}(X) = p(1 - p)$ .

**Memory hook:** Bernoulli = ONE yes/no trial.

### 5. Binomial

Binomial counts successes across  $n$  repeated Bernoulli trials with fixed  $n$ , common success probability  $p$ , and independent trials.

$P(X = k) = C(n, k)p^k(1 - p)^{n - k}$ . Mean =  $np$ . Variance =  $np(1 - p)$ .

Example: if  $p=0.2$  and  $n=5$ ,  $X$  can be 0,1,2,3,4,5 successes.

**Memory hook:** Bernoulli = one trial. Binomial = many trials + count successes.

## 6. Expected Value

Expected value is the long-run average outcome:  $E[X] = \sum xP(X=x)$  for a discrete variable.

Example: ■200 with probability 0.5 and ■0 with probability 0.5 gives  $E[X] = \blacksquare 100$ .

Expected value is not necessarily an outcome in one trial; it is the average over many repetitions.

**Memory hook:** Expected value = long-run average.

## 7. Variance & Standard Deviation

Variance measures spread around the mean:  $\text{Var}(X) = E[(X-\mu)^2]$ . Standard deviation is  $\sigma = \sqrt{\text{Var}(X)}$ .

If variance=16, SD=4. If mean=100 and SD=4: 1 SD below=96, 2 below=92, 1 above=104, 2 above=108.

Squaring prevents positive and negative deviations from cancelling and makes extreme deviations count more.

**Memory hook:** Variance = spread squared. SD = spread in original units.

## 8. Covariance & Correlation

Covariance tells whether two variables tend to move together: positive=same direction; negative=opposite directions.

Correlation standardizes linear association to  $-1$  through  $+1$ .  $+1$  is perfect positive linear correlation;  $-1$  perfect negative;  $0$  means no linear correlation.

Correlation  $0$  does NOT prove there is no relationship: a nonlinear relationship can have zero correlation.

**Memory hook:** Covariance = direction together. Correlation = standardized linear relationship.

## 9. Normal Distribution

Normal distribution is the symmetric bell curve. Mean=median=mode.

The 68–95–99.7 rule says approximately 68% is within  $\pm 1$  SD, 95% within  $\pm 2$  SD, and 99.7% within  $\pm 3$  SD.

SD controls width/spread; mean is the center.

**Memory hook:** Normal = symmetric bell curve.

## 10. Z-Score

$z = (x - \mu) / \sigma$ . It tells how many standard deviations an observation is from the mean.

Example:  $\mu=100$ ,  $\sigma=4$ ,  $x=108 \rightarrow z=2$ , so 108 is 2 SD above the mean.  $x=88$  gives  $z=-3$ .

Positive  $z$ =above mean; negative  $z$ =below;  $z=0$ =at mean.

**Memory hook:** Z-score = distance from mean measured in SD units.

## 11. Central Limit Theorem (CLT)

Imagine 10,000 students. Repeatedly take samples of 100 and calculate each sample mean: 168, 171, 169, 170, ... The collection of means forms a sampling distribution.

Under appropriate conditions, CLT says the distribution of sample means becomes approximately Normal as sample size becomes sufficiently large, even if the original population is not Normal.

The sampling distribution of the mean has mean  $\mu$  and standard error  $\sigma/\sqrt{n}$ .

Example:  $\mu=50$ ,  $\sigma=20$ ,  $n=100 \rightarrow SE=2$ . If  $n=400 \rightarrow SE=1$ .

Important: CLT does NOT say the original observations become Normal.

**Memory hook:** CLT = samples  $\rightarrow$  means  $\rightarrow$  repeat  $\rightarrow$  means become approximately bell-shaped.

## 12. Population vs Sample

Population = the entire group you care about. Sample = the smaller group you observe.

Example: 1,000,000 customers = population; 5,000 selected customers = sample.

Population mean  $\mu$  is a parameter. Sample mean  $\bar{x}$  is a statistic. If sample mean = 2,400 and true population mean = 2,350, 2,400 is a statistic and 2,350 is a parameter.

Samples differ from populations because of sampling variation; biased sampling can create systematic error.

**Memory hook:** P  $\rightarrow$  Parameter  $\rightarrow$  Population. S  $\rightarrow$  Statistic  $\rightarrow$  Sample.

## 13. Standard Error

Standard error measures how much a statistic such as a sample mean varies across repeated random samples.

For a sample mean,  $SE = \sigma/\sqrt{n}$  (or approximately  $s/\sqrt{n}$  when  $\sigma$  is unknown).

Example:  $\sigma=500$  and  $n=5,000 \rightarrow SE \approx 7.07$ . If  $n=20,000 \rightarrow SE \approx 3.54$ .

SE is not the exact error of one sample mean; it is the standard deviation of the estimator's sampling distribution.

**Memory hook:** SE = how much your estimate bounces around across repeated samples.

## 14. Law of Large Numbers (LLN)

LLN says that as the number of observations/trials becomes very large, the sample average tends to approach the true expected value.

Fair coin: 7 heads in 10 flips = 70%, but with 10,000 flips the observed proportion will typically be close to 50%.

LLN focuses on the average approaching truth; CLT focuses on the shape of the distribution of repeated sample means.

**Memory hook:** LLN = more observations  $\rightarrow$  average becomes more stable and approaches truth.

## 15. Skewness

Skewness describes asymmetry and the direction of the tail.

Right/positive skew: long right tail; high values pull the mean right, so generally mean > median.

Left/negative skew: long left tail; low values pull the mean left, so generally mean < median.

For heavily skewed data, median is often better for the typical observation because it is less affected by extremes.

**Memory hook:** The mean follows the tail.

## 16. Outliers

An outlier is unusually far from the rest of the data. It is unusual, not automatically wrong.

Example: 10,11,12,13,100 → mean=29.2 but median=12. The mean is heavily affected by 100.

Variance and SD are sensitive because deviations are squared. A deviation of 88 contributes  $88^2=7,744$  to a squared-deviation calculation.

Do not automatically delete outliers. In fraud detection, extreme transactions may be legitimate or may be the signal you need.

A common rule of thumb is  $|z|>3$  as a potential outlier, but it is not universal.

**Memory hook:** Outlier ≠ bad data. Investigate before removing.

## 17. Percentiles, Quartiles & IQR

Percentile describes position. The 90th percentile means approximately 90% of observations are at or below that value.

$Q1=P25$ ,  $Q2=P50=\text{median}$ ,  $Q3=P75$ .  $IQR=Q3-Q1$  and represents the middle 50% spread.

Common fences: lower= $Q1-1.5 \times IQR$ ; upper= $Q3+1.5 \times IQR$ .

Example:  $Q1=10$ ,  $Q2=20$ ,  $Q3=30$  →  $IQR=20$ ; upper fence=60; lower fence=-20. Therefore 65 is a potential outlier.

Real-world: P95 delivery time=42 minutes means approximately 95% of deliveries take 42 minutes or less.

**Memory hook:** Percentile = position.  $IQR = Q3-Q1$  = middle-50% spread.

## 18. PMF — Probability Mass Function

PMF is for discrete random variables and answers: “What is the probability of this exact value?”

Example: number of complaints:  $P(X=0)=0.50$ ,  $P(X=1)=0.30$ ,  $P(X=2)=0.15$ ,  $P(X=3)=0.05$ .

So  $P(X=2)=0.15$  means 15% probability of exactly 2 complaints.

Bernoulli, Binomial and many count distributions are naturally described with PMFs.

**Memory hook:** PMF = DISCRETE + EXACTLY.

## 19. PDF — Probability Density Function

PDF is for continuous random variables. It describes probability density, not probability at an exact point.

For a continuous variable,  $P(X=42)=0$ . Probability comes from an interval/area, such as  $P(40<X \leq 42)$ .

The height of a PDF can be greater than 1. The total area under the PDF must equal 1.

Think of PDF like density: it tells how concentrated probability is around values.

**Memory hook:** PDF = CONTINUOUS + DENSITY. Area under the curve = probability.

## 20. CDF — Cumulative Distribution Function

CDF answers: “What is the probability that X is less than or equal to this value?”  $F(x)=P(X \leq x)$ .

CDF works for both discrete and continuous variables and never decreases.

If  $F(30)=0.70$  for delivery time, 70% of deliveries take 30 minutes or less.

If P95 latency is 200 ms, approximately  $F(200)=0.95$ .

**Memory hook:** CDF = BOTH discrete and continuous + UP TO.

## 21. PMF vs PDF vs CDF — Interview Cheat Sheet

| Concept | Variable   | Question   | Example        |
|---------|------------|------------|----------------|
| PMF     | Discrete   | Exactly?   | $P(X=3)$       |
| PDF     | Continuous | How dense? | $f(30)$        |
| CDF     | Both       | Up to?     | $P(X \leq 30)$ |

**Three words:** PMF = EXACTLY • PDF = DENSITY • CDF = UP TO

## 22. Level 1 Interview Exercises

1. A user has a 20% chance of opening a notification. What is the probability they do not open it?
2. Five independent users each have a 10% chance of purchase. How would you calculate the probability of at least one purchase?
3. Out of 100 customers, 40 bought a product and 30 of those buyers are male. Find  $P(\text{Male} \mid \text{Bought})$ .
4. A transaction is flagged. For “probability it is actually fraud,” should the denominator be all transactions or all flagged transactions?
5. What distribution describes one yes/no trial? What distribution counts successes over 20 independent trials?
6. A game pays \$200 with probability 0.5 and \$0 otherwise. Find expected earning.
7. Mean=100, SD=4. Find the z-score of 108.
8. What becomes approximately Normal under CLT: original observations or distribution of sample means?
9. Why does the observed proportion of heads tend toward 50% as fair-coin flips increase?
10. A salary distribution has a long right tail. Which is generally larger: mean or median?
11. For 10,11,12,13,100, which is more affected by 100: mean or median?
12.  $Q_1=10$  and  $Q_3=30$ . Find IQR and the upper outlier fence.
13. For a continuous variable, can  $P(X=42)$  be positive under the usual continuous model?
14. If P95 delivery time is 42 minutes, approximately what is  $P(X \leq 42)$ ?

## 23. Answer Key + Reasoning

1.  $80\% = 1 - 0.20$ .
2.  $1 - 0.9^5$ . The complement is easier than adding cases for 1,2,3,4,5 purchases.
3.  $30/40 = 0.75$  because “given Bought” restricts the world to the 40 buyers.
4. All flagged transactions. “Given flagged” means the denominator is the flagged world.
5. Bernoulli for one trial; Binomial for successes across fixed repeated independent trials.
6.  $\$100 = 0.5 \times 200 + 0.5 \times 0$ .
7.  $z = (108 - 100) / 4 = 2$ .
8. The distribution of sample means, not the original observations.
9. LLN: with many trials, the sample proportion tends toward the true expected probability.
10. Mean > median because the mean follows the right tail.
11. Mean; median is much more robust to the extreme value.
12. IQR=20; upper fence= $30 + 1.5 \times 20 = 60$ .
13. No. For a continuous variable,  $P(X=42)=0$ ; an interval can have positive probability.
14. Approximately 95%.

## 24. Level 1 One-Page Mental Map

**Probability** → likelihood

**Complement** → opposite =  $1 - P(A)$

**Conditional** → shrink world to “given”  
**Bayes** → update belief using evidence  
**Bernoulli** → one yes/no  
**Binomial** → many trials + count successes  
**Expected value** → long-run average  
**Variance / SD** → spread  
**Covariance / correlation** → variables moving together  
**Normal** → symmetric bell curve  
**Z-score** → SDs from mean  
**Population / sample** → whole group / observed subset  
**Parameter / statistic** → population number / sample number  
**LLN** → more observations → average approaches truth  
**CLT** → repeated sample means → approximately Normal  
**SE** → spread of an estimator across samples  
**Skewness** → direction of tail  
**Outlier** → unusually far observation  
**Percentile** → position  
**IQR** → Q3–Q1, middle 50% spread  
**PMF** → discrete + exactly  
**PDF** → continuous + density  
**CDF** → both + up to

## 25. FAANG / MAANG Interview Mindset

- Do not start with a formula. First identify the situation and what is being asked.
- Always understand the denominator in conditional probability and classifier questions.
- Know mean vs median under skew/outliers, and variance vs SD.
- Be able to explain CLT vs LLN in one sentence each.
- PMF/PDF/CDF should be automatic: EXACTLY / DENSITY / UP TO.
- Connect statistics to product problems: conversion, fraud, spam, latency, customer spending and experiments.
- Before applying a method, ask: discrete or continuous? independent or dependent? representative or biased? outlier or genuine rare event?

## Next: Level 2

After you can solve the exercises without looking at the answers, move to Level 2: sampling distributions, confidence intervals, point estimates, margin of error and statistical inference.